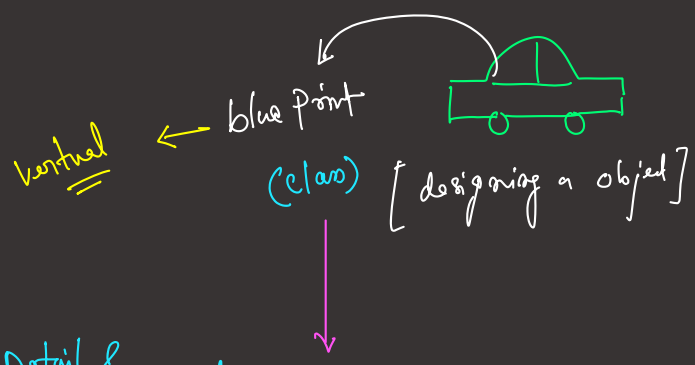


Oops

Object Oriented Programming.



Principles of Oops

1. Abstraction [hiding internal detail & showing only required things] Real world Entity
2. Encapsulation [everything in a single box] (Object)
3. Inheritance [old feature with some extra new feature]
4. Polymorphism [Multiple things by a single name]

class Car {

① Properties { int windows
int wheels
int seats } Private

② behaviour [method] Public
↳ it will utilized the properties

1. Specialization [Inheritance]
2. Generalization [Polymorphism]

* toString()

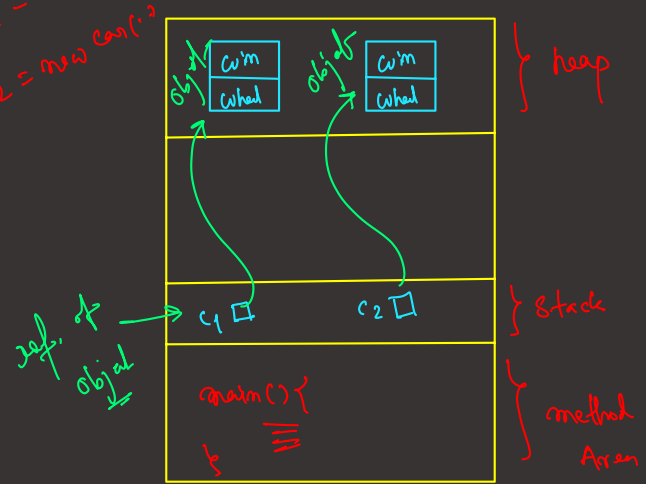
ArrayList arr = new ArrayList(1, 2, 3, 4, 5)

arr → [1 | 2 | 3 | 4 | 5]

System.out.println(arr) →
1, 2, 3, 4, 5

Car c1 = new Car(4, 4)

Car c2 = new Car(2)

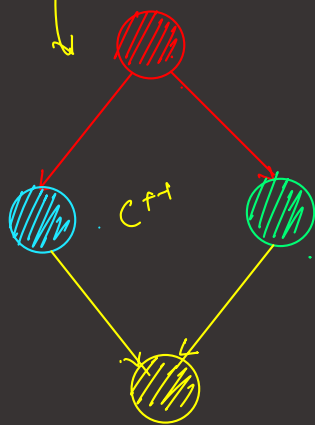


java Access specifier

	default	private	protected	public
same class	yes	yes	yes	yes
same package subclass	yes	no	yes	yes
same package non-subclass	yes	no	yes	yes
different package subclass	no	no	yes	yes
different package non-subclass	no	no	no	yes

2 Inheritance

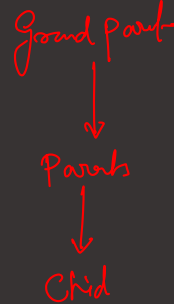
Diamond Problem



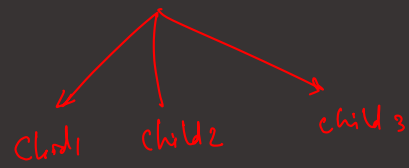
Single Inheritance



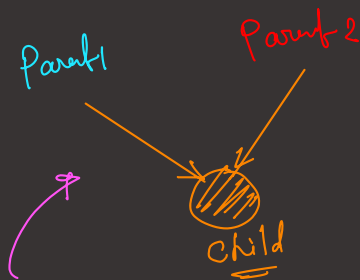
Hierarchical Inheritance



Parents [Multi level inheritance]



Multiple Inheritance



not in java (class)

using interfaces it is possible

hybrid Inheritance

